

ActInf Livestream #028.0 ~ “Towards a computational phenomenology of mental action”

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hello and welcome to the active inference lab it's actin flab live stream number 28.0 on september 6 2021 we're going to be talking about the paper towards a computational phenomenology of mental action modeling meta awareness and attentional control with deep parametric active inference welcome to the active inference lab we are a participatory online lab that is communicating learning and practicing applied active inference you can find us at the links on this page this is a recorded and archived live stream so please provide us with feedback so that we can improve on our work all backgrounds and perspectives are welcome here and we'll be following good video etiquette for live streams thanks a ton blue and steven and dean nice fun to have a zero crew together here you'll see a slightly updated version of the upcoming streams calendar which you can get at from this short link here uh september 7th and 14th we're going to be discussing this paper we're going to be discussing in this one and then we don't have a paper for 29 and then we have some papers set with authors for all of october also check out the tabs to see other series other than the tuesday paper centric walkthroughs and of course anyone's welcome to co-organize an event or to be the guest for one of these types of streams all right so today in 28.0 we're going to be just learning and discussing this paper that was already mentioned by sandvik smith hesp mataut fristen lutz and ramstead out only a few weeks ago which is kind of cool to see and there's going to be a time to talk about it's a super interesting paper and if you want to participate in 28.1 or two you should get in touch otherwise if it's after that time then check out those videos and maybe find another one to participate in but i'm just looking forward to hearing what you all were excited about by way of introduction just want to hear from everybody and what was exciting to them about this paper or made them want to join for the dot zero um if i go first uh daniel yeah thanks um i'm really excited to or interested to see them moving beyond a computational approach to neuro phenomenology which i've seen before and into this broader idea of what they're calling computational phenomenology which wears a bolder claim and has more

relevance potentially to other applications which i'm interested in and applied active inference context so you know we'll see how that bold claim can be uh applied okay so for me i thought it was like ultra meta um the paper really got into the hierarchical um levels of organization which i'm always excited to see and talk about uh for me it was the fact that i am doing some writing right now and i am interested in the idea that there is a way of putting this into some sort of a geometric relationship as well as the computational relationship cool okay uh the paper is let me just re-frame it is towards computational phenomenology of mental action modeling meta awareness and attentional control with deep parametric active inference in neuroscience of consciousness published um about two weeks ago 27th of august does someone want to read the three aims of the paper sure i'll go ahead the three aims are to explicitly model meta awareness to account for the computational consequences of meta awareness on cognitive control and lastly to provide a computational account of mental covert action and then maybe the second part sure formally speaking the crucial thing to note about these cognitive processes is that they are about other such processes the deployment and control of attentional states and processes for instance are quintessentially a higher level ability such states and processes monitor and monitor and modulate other typically lower level perceptual states third order higher level processes in turn oversee these attentional processes we can be aware of our attention being grabbed and make a deliberate decision to focus our attention elsewhere it would seem then that effective self-regulation of attention depends on the ability to access evaluate and control the quality of these attentional processes themselves in the same way that attentional processes are necessary to consciously access assess and control lower level perceptual states interesting areas it's just interesting to see that the kind of framework that helps a robot navigate now it's being used in a little bit of a different way like nesting within ourself and coming closer to our experience and other things which often are said to be unable to be directly observed so it's quite a cool topic uh dean or stephen any thoughts or do either of you want to read the first abstract i got a quick thought before we get to the abstract um in the introduction of the paper the i just wanted to read this really quickly it's in the paper but computational terms the central place of confidence and reliability and effective self-regulation speaks to the key role of precision or inverse variants of implicit beliefs and then they talk about probabilistic and posterior-based beliefs that describe a probability distribution over some latent quality but this was the thing that i found really interesting bayesian beliefs about bayesian beliefs can in certain situations become propositional and the way that they are estimated optimized and controlled precisions in a

nutshell quantify our confidence and our beliefs say how confident we are about what we know about states of the world about their relationship to our observations or about how these states change over time being able to place numbers around that as soon as i read that and and saw it in relationship to a proposition i found that really interesting and one it was really curious to see now so how is that thread gonna play out now it wasn't that wasn't part of the three things that were in in the previous slide but i think it was really important and something that was it has to be a name or a claim of the paper because it's pretty fascinating good call it's almost like a background assumption whether uh post hoc justified by empirical appeal or not that precision is what matters which is kind of the other side of the coin from uncertainty like high precision means low uncertainty and vice versa and sometimes they're framed as so i think totally agree that is a claim of paper stephen want to go for abstract one [Music] okay meta awareness refers to the capacity to explicitly notice the current content of consciousness and has been identified as a key component for the successful control of cognitive states such as the deliberative direction of attention this paper proposes a formal model of using hierarchical active inference to do so we cast mental action as policy selection over higher level and at a further hierarchical level to model meta awareness states that modulate the expected confidence or precision in the mapping between observations and hidden cognitive states yeah there's a lot of terms some of which we're gonna go into in the keyword section but pretty distilled down description of their contribution and we're gonna unpack it more abstract two uh we simulate the example of mind wandering and its regulation during a task involving sustained selective attention on a perceptual object this provides a computational case study for an inferential architecture that is apt to enable the emergence of these central components of human phenomenology namely the ability to access and control we propose that this approach can be generalized to other cognitive states and hence this paper provides the first steps towards the development of a computational phenomenology of mental action and more broadly of our ability to monitor and control our own cognitive states future steps of this work will focus on fitting the model with qualitative behavioral and neural data cool topics any thoughts or we can move on first first steps that's the emphasis here just first steps yep i'm pretty excited to get to this um the generalized model of mental action because it's like almost like the new tool enables asking about phenomena that we already know and then it reveals a whole new state space of kind of then the model feeding back to the kinds of regimes of attention that we design for ourself or niche modifications or information inputs roadmap so this paper has an interesting layout it is just an introduction that brings up the non-active

inference sections related to the scientific study of mental action and about how it's been computationally modeled outside of active inference then there's a methodology paper that first has more of a model building uh narrative and then the empirical results from a simulation that has a few different layers of detail layered on it it's not just one single model there's a few little variants that help get out which parts the model are contributing which behavior and why then they have a more general discussion and give some directions for future research and that's where some of these big claims about computational phenomenology go and then there's a conclusion so any thoughts on the roadmap yeah i was gonna i think that this gives a good sense of the way that they're trying to enter into this space looking at the opacity and transparency of phenomenology and uh trying to sort of bring back that neuro phenomenology into this wider phenomenology uh so by by sort of looking at the experience now the question that sort of sort of comes up with this is because obviously a lot of the work here with um ramstead and others has been around in body cognition and stuff like that so they're sort of going into this idea of trying to set it up so that they can target phenomenology while still keeping it around mental states and cognition so there's an interest in uh i don't know ontological dance they're trying to find a way through here which i think is kind of interesting any other comments on the roadmap so the keywords were free energy principle active inference neurophenomenology metacognition opacity transparency focused attention mind wandering and mindfulness so perhaps blue with fvp sure this slide i pulled out of the live stream 15.0 so the free energy principle is based on three aspects the observation of self organization which brings us to the second aspect which is that living organisms can be described as stochastic dynamical systems possessing attractors and then lastly the states in which the self-organizing system is at a point in time can be identified by the interactive role they play within the multi-level self-organization scheme and then this is related to in the slide from 14.0 multi-scale systems ensembles info thermodynamics and variational bayesian inference which i thought was applicable here as we're going through this hierarchical awareness meta awareness cool it yeah stephen and then i'll have to die yeah well yeah i think this is good good to characterize this and also i know that um maxwell mentioned how that multi-level graph he's kind of gone back on it a bit but it does show that whole question is a really good diagram to show both that there is this multi-level nested sort of multi-scale thing going on and also it's kind of a contested it's a contested kind of evolving part of the whole field because it's uh people are sort of putting out things and then they're maybe backtracking a little bit and so i think um yeah i think it's quite a good thing to bring up definitely uh not a

settled area that's why it's so fun to just have these discussions on the topic because even the paper is not the end result it was first steps we're going to see in this discussion the multi-scale system is almost like psychologically nested or mentally within one agent whereas previously we've been talking a lot about multi-scale systems in the context of you know a bunch of ants making the colony a bunch of communicating individuals making the community so it's like that's like multi-scale systems out and getting spatially bigger and this is kind of getting more ephemeral but also having this very interesting influence on the lower level models so for active inference the key word we can look at what the authors wrote because it's always cool to see just how did they define this how did they frame what active inference inferences for this paper and they wrote that at the root of these advances is a biologically plausible neurocognitive and behavioral modeling framework called active inference active inference provides us with a bayesian mechanics that is the mechanics of knowledge driven action and active prep perception which explains from first principles how autonomous agents can thrive within their ever-changing environments active inference descends from and is closely related to bayesian brain theory and predictive coding and then this green one i thought was very interesting in how they said it was active inference cast perception learning and action as essentially being in the same game that of gathering evidence for the model that underwrites the existence of the agent so just on the diagram that we've seen with the four different kinds of nodes with the blanket states of in and out going in the internal and external states like to say that the three that constitute the organism let's say are all in the same game just bring sort of a unity of action and inference to how we think about the very least other systems that already have this structure like input output systems which some people speculate describe all classes of systems or computational systems so it's just like the way that they made that clear of active inference without tying it to a specific model implementation i thought was very well said stephen and also this does give a flexibility as knowledge evolves because as mark soames has been talking about the the way attention may be moved around more at the base of the brain in it using more kind of attentional biological you know inactive questions that wouldn't necessarily be a problem for this because this this modeling isn't completely tied to some sort of frontal cortex model of thought it's kind of it's broader so i think that's that's quite interesting and to see how they're called using the term behavioral modeling they framework and they sort of yeah the way they positioned it because they haven't gone the whole way and talked about biological and gone down the full multi-scale biology route they've kept it kind of in the brain but they've brought in that behavioral element which i think neurophenomenology

maybe doesn't do as much good segway blue want to describe this one sure so neurophenomenology in the words of the authors neurophenomenology has been described as the search for generative passages between the neurobiological phenomenological and mathematical levels of description going beyond the description of mirror isomorphisms between these levels toward mutual epistemological and methodological methodological accountability and cross-fertilization neurophenomenology aims to bridge the gap between these kinds of data via the formal metaphysically neutral levels of the description provided by mathematics and computational work and i pulled this cartoon out of this paper because i thought it was kind of a neat um perspective and visual under phenomenology so it talks about the biological structures as like pre-determining the world of the organism and then neurophenomenology as a proxies between the first person account of the the organism has and the neural descriptions of these structures so that that was a kind of the point of this nice find very interesting figure looks kind of like a sci-fi human novel but it's kind of cool okay um to talk about one specific kind of neurophenomenology like a first person experience that is desired to be modeled potentially even in a computational direction here they're focusing on this metacognition aspect so just the yellow parts they wrote redefine meta awareness in this context as the ability to explicitly notice the current content of a conscious episode for our purposes as discussed in the text we define states of meta awareness as those third order which we'll get to states of the system that are about and enable the opacity of second order attentional states metacognitive states thus function analogously to attentional states which enable the opacity of first order perceptual states so it's going to be a three layer nesting doll the role that the second model plays over the first is going to be analogous to the role that the third model plays over the second but that doesn't mean its dynamics are going to play out in the exact same way and then this last part was very interesting like mental action refers to goal-directed internal behavior without necessarily having a direct consequence on the external environment e.g as though muscular a paradomatic example of mental action is covert attention so what an interesting topic you know the whole think of x or don't think of x i won't play any language games on the live stream but many of those exist and they have to do with inducing mental action so that's just so interesting that that's what this framework is going to be exploring stephen go for it yeah and without you know putting too much pressure on the paper to go further but is it that question of skill performance and the phenomenology of being in the flow or being in a state of sense making it'd be interesting how that neurobiological phenomenology phenomenological passages um can extend out into some of that kind of work

um which is obviously highly complex but uh then you know this might give us clues so that's good yep cool and then oh this is just i found this link uh trailblazing ants show hints of metacognition when seeking food and um recent research has shown that bees when faced with a difficult task simply opt out of doing it a behavior that may be interpreted as a form of metacognition at least computationally now it seems that ants might be capable of a similar mental feat despite their tiny simple brain it's like what computational attributes are we going to assign to oh but that's just how bark is or that's just how ants are versus this variable is the one that we're experiencing because of this so that's just kind of a topic that's going to come up probably multiple times okay how about transparency and opacity what does it mean in someone's words or reading here all right can i bring out a little different angle to it i mean i understand that how the contrast of transparency and opacity works but when i was bringing when i was trying to figure this out to help somebody who was walking into a situation oriented to a lot of the data around them that they would otherwise be overwhelmed with one of the things we talked about was the difference between liminality and just getting a generalized sense of the orientation space and then limitality sort of filling in the details and getting and understanding where the edges are so that you don't bump into them and hurt yourself and so i get the transparency and opacity piece and how it's explained in the paper but for a person who's trying to move from the generalized to the more specific i think i think it's it's the opacity piece speaks clearly to the difference between that and what something that's clear but i also think that the details get filled in when you can have an understanding what the limits are and not just limitly understand that it's now in your in your field of awareness which is how they're describing this which is your where's your focus where's your angst i put this goldfish meme it's the story about the goldfish and what is water and the elder one is aware of water and then the younger ones are unaware of the water that they swim in cough cough and then it's kind of like the two mental extremes like water is opaque you can think of the water being like cloudy to this fish maybe it has a different vision so then it can see that there's a drop-off in vision it infers that it's in opaque water so water is something that it can ask about the conditions of whereas the small fish in this example it's clear it's like transparent so they're seeing through it ergo it's transparent to them so they're imperceptible so the green ones are accessible to the system you're aware of it it's like water that has an opaque tint and then the transparent means like you just can't differentiate it experientially and that's a philosophical framework by i think metzinger which has probably been influential right yeah highly influential i think uh increasingly so um i think i think that i think this could be pushed

a bit further because i think it's okay to state this as a almost a philosophical piece but bringing in stuff like mental space psychology which hasn't really been incorporated in this area but you know these subconscious um sort of opaque or transparent awarenesses are part of our could be part of a deeper way of knowing if we assume that thinking is through our senses if sense if actually the process of the brain is not all in the box but it's recapitulating stuff through the sensory world around us even in the subconscious that would tie in quite well with this but also give a bit more of a computational grounding the other thing too daniel if you'd have written the words opaque and transparent in a white colored in a white colored font with that white background they'd still be opaque but we wouldn't see them on the back on the white background and so that's again where nice good call in order to be able to bring the idea forward it isn't just a comparison of transparency to something that actually has a a sense of tangibility um there's this other p part of it that we will pick it up because there is a contrast there is some differentiation and doesn't always just have to be the something's clear and something has depth thank you steven yeah that's a really good point and actually that can extend as well to the size of the space you're in so if you start to imagine the relationships or these kind of subconscious senses of how close or far certain ideas are that changes depending on the size of the actual room you're in and i imagine the different colors of the room the light in the room as well might affect i've only thought about this before it might affect the transparent states as well because there may be some sort of calibration going on you know so it might be quite profound to be in a blue room in terms of what your transparent end up recapitulating so that's interesting we'll come back to i think uh niche modification and design of regime of attention who made this slide or what did it mean to anyone so this was mine um the focused action i thought was like an interesting keyword the authors didn't use it once if you search the paper and then as a google search you search for focused action and you get all kinds of life coaching which is like whoa okay i don't think that that's maybe the the intention here so um this is just the typical active inference loop where you have internal states perception external states and action that are all connected and and partitioned by a markup blanket so in the i'm extrapolating based on what i think that the authors mean low precision is unfocused and and it's low precision also in terms of perception because like perception is also when you like register you just saw something they talked about the oddball and like you know you're seeing an image seeing an image seeing an image and then all of a sudden the image is different and it's like oh i wasn't even paying attention to what i was seeing at all so it's that perception can also have this kind of unfocused quality so so i related this to like a low

precision kind of perception or action as unfocused and where it's high precision where you're focused on what you should be focused on that's that's focused thanks just one note on that is under the reward paradigm plus minus valence avoid approach sort of forward and backward linear mindset we can say oh well not just do we diagnose happiness or sadness or on a linear scale but it's a goal and so just what are the aspects of our experience or of just the outcomes of systems that can be said to be related to focus and attention and action so how about just mind wandering since it's related topic sure this is um these are different papers and so my pondering has been explored in several different contexts but specifically like the figure on the left is um by wanda hassenkamp and she studies uh she's at the mind life institute and so studies directly meditation and the effects on the brain and here it just kind of shows the different um brain regions that that are activated during meditation or or also it's like where you have awareness and then your your mind is wandering you become aware that your mind is wandering and then you shift your mindset back to like oh i better be paying attention to my meditation for example focusing on my breath or whatever and then you have a state of focus again um and this kind of contrasting neural model of mind wandering this is a 2016 paper and here they they discuss this in a very different way so when i think about mind wandering it's easy to think about in the context of um meditation but here they describe like okay if you're focused on a task that's on task and then mind wandering is actually pursuit of internal goals which i thought was interesting like you're still focused but you're focused on your shopping list or whatever like internal thing that you want to focus on even though you're like this is like mind wandering and then they call this off focus is epistemic exploration or this is like the exploratory like where your mind what i think of as mind wandering they they call this off focus and so i thought that that was kind of a different paradigm and not something that i'm used to hearing like where you're just where you catch yourself like why was i thinking about red balloons and and you know frogs with three legs like where you just find yourself in this intellectual state like what am i thinking about or why was i even thinking that whereas mind wandering is like you really are thinking about like some something else that's like you should be thinking about but not it's not exploratory which i thought was kind of interesting well the things i'd say about that on the bottom right first it doesn't look like there's a transition from the bottom left to the bottom right so it could just be that the dynamics the tractor dynamics of this dynamical system are such that when you just down project these massive data sets of neural imaging you get certain transition dynamics which is totally the case like you're sleepy before you fall asleep and then you know different people accelerate or have different

dynamics but there is kind of a trajectory so that was one point was these could be very nuanced states and then the space and the transitions that exist or don't can be important and the second thing is that the y-axis has to do with some neural activity some neural marker where it's low in the bottom two and high in the middle one but then the x-axis is behavioral results so the idea would be that there might be some correlation between like wow it was like they did 30 seconds of really good task and then there was 15 seconds where they totally didn't do the task or was variable and then if you found a neural change that was like immediately preceding that or after it or during it that's the kind of cross data set connections that this research is getting towards but instead of just putting this into the broad framework of dynamical systems and statistics now we have a process theory like active inference to explore some of the dynamics that give rise to those patterns so blue and then d so interestingly there is a correspondence um so there it does map what was found and i had to like read into the paper to kind of find this out but the mind wandering in the left image does map actually to the mind wandering in the image on the right or or the brain regions are are super similar so i mean i i read to just make sure that there was some cohesion there and there is so so it's like interesting to me to think about this off task like what what is like off task versus mind wandering and and what like for me if you're meditating and you're not like focused on the the thing like you're off task but it's i don't know super interesting i just wanted to clarify that one point though that there's there is a cohesion there thanks blue dean yeah i used to have to work with people who were at a real i gotta i gotta be polite because i'm being recorded they had they had a real joy out of out of um looking at time on task this was something that was a something that they found really really important and so if you were task designing you were supposed to be able to create something in your in your design scheme that would keep people focused all the time and in the meantime there's a whole bunch of interesting papers that have come out in terms of creativity and tangential thinking and mind wandering as an actual gift for taking what's in the task and making it your own like if you're mind wandering you're not it's not a disconnected thought that you're having and unless you're really being distracted by the video game or the music or whatever it's it's usually something tangential to the task at hand and it's it's essentially an embodying exercise so to i i think this is good stuff just because i think lots of times mind wandering if you're if you're focused on how many minutes was a person focused or concentrated is seen as a a negative when it from creative stance it's a really important thing to do thank you dean and then stephen yeah i think it's it's useful to see a transition some sort of transition state between the regime of attention where you're on task

and a regime of attention where your mind wondering so i think that that opens up questions of how things you know liminal spaces between states um the need to like if you're always coming back on task yeah it would and the you're not getting beyond that state of losing focus that that could be a problem i mean it can be a problem if the mind wandering doesn't get back on task as well i suppose but uh so i think that's um that would that would make a lot of sense i think how about um mindfulness so just the yellow part and then blue or anyone can give a thought on what they added um or for the purpose of this work they're going to focus on regulatory cognitive control strategies in which an agent seeks to control their mental states by becoming aware of the state as a cognitive process as opposed to regulation strategies where one remains engrossed in the contents of the state this style of regulation is central in mindfulness related intervention where patients learn to change their relation to thoughts and emotions rather than change the thoughts and emotions themselves so this is pretty interesting so cognitive process is like this is a thought i'm having a thought there's many many praxis many knowledge traditions built around this which you can mention or have people with experience talk about and the alternative is like when you feel x feel the x and be the x that's sort of the opposite q so we're thinking like mental action cues for action like cues for a sport dean's favorite topic and then there's cues for mental action that are verbal or have other modes of being delivered and what are the cues for mental action that leads to different regimes of attention like steven said and then how are the measurements that we're going to make from different modalities going to inform that study and the kind of modeling that integrates across them which we'll get to as well stephen yeah this is a useful piece here it's good to distinguish between i mean often with children for instance there's now more emphasis on mindfulness there's that big push to just get them busy you know make them do sports make them do this make them do the other so they're so busy with something they can't think about anything else but actually saying to them okay you're going to be in a space where you could think of other things and you're going to work on how to not focus on other things has some benefits i think there's some slight downsides of that if someone's got trauma because then it opens up the space for those things to be coming into play so i suppose i think this is really helpful to to see where they're going and it'll be interesting to know um how this plays out because there's always a danger if someone tries to scale this type of work because it's also easier to use the insights from this type of work to try and capture someone's attention at scale or capture many people's attention by using some of the levers here as much as it is for the person on the inside to be able to resist all the distractions and the the flashes of light that

they keep getting exposed to in their daily life thanks steven blue yeah kind of i want to just follow what uh steven said so um i don't know that the children or or even like us adults being constantly exposed to flashing blue light like the smartphone like instant dopamine addiction you know fulfillment thing is necessarily the opposite of mindfulness i i think that that that feeling is maybe the elimination of boredom which you can have mindfulness with boredom or mindfulness while doing nothing and mindfulness while not doing nothing so you can you can have actually not sitting in mindfulness meditation but but retain through this is called like you know meditation practice the meditation off the mat right so like there's there's the conscious placement of the mind during the meditation session but then when it's over you carry that into your daily life by constantly like asking yourself like where is your mind which is like this that's the image i put in about the old pixie song where is my mind so so there is the ability to carry mindfulness even if you're very busy you can still practice mindfulness in your daily life without taking time to sit on the cushion so i think we are starving our children of boredom which maybe is i i'm more focused on or more worried about a lack of creativity because i think that that boredom process where your mind is really wandering to red balloons and frogs with three legs like that is where um the creative the creative nature tends to bubble up awesome and i won't mention the person's name but i hope that they've emailed us um they have emailed us so i hope they talk more about it but they've related these aspects of active inference to the yoga of action karma yoga and and um basically like the way that action exists in the world so i hope stop what my knowledge area is so look forward to learning about that though so before we jump into the paper this has been a nice long background because there's a lot to say and it's pretty interesting look at this fun april first april fool's day with steven and blue and i but they write in this paper that the key technical innovation presented in the paper is an extension of the active inference framework derived from the work of hesp at all 2021 which was the paper that we discussed in live stream number 19. if you want kind of the kernel of what this discussion is about then check out number 19.0 and 0.1.2 and we were talking then about affect and about the way that that made deep agents so listen there to learn more and now we're going to be doing a few of the similar things seeing some similar figures so it's a good background listen okay figure one does anyone want to give a first description it says in the in the caption a probabilistic graphical model showing a basic generative model for moment to moment perception okay so the observation o is going to be used across different layers but here is the model of perception so we can debate whether it's realism or not later but this is kind of like the retina or the sensor of the camera oh it's like did a photon hit it or not

this is an observable that you measure with a sensor s at each level is going to be a hidden state so called because it's not directly observed but it's inferred from observations through a which is a mapping so like this could be day and night and this could be photon hitting versus no photon hitting and then you can imagine the mapping there but we'll see how they nest and get more complex than that and then d is the initiator it's a prior overridden stage which turns out to be important because as you compute the likelihood of different hidden states being the case um moment to moment or through time you need to start with an idea of essentially how likely certain states are and how closely they map to observations that's the kernel of the bayesian modeling that we're going to elaborate a lot like multiple successive times but this is the underlying system that can be thought of like photons hitting the receptor and a mapping to underlying as influenced at least initially by a prior okay any comments on one or we can go to the next generalization and two okay so two is a bayes graph showing a basic generative model of perception with precision so it has the same terms and you know d s a o as above d s a o but now there's a new node and each shape just at a first pass represents some value that can be taken by a variable and then an edge that connects two is like a relationship between them and there's different kinds of nodes there's different kinds of edges slightly in how they influence each other but they just mean that they influence each other and that's important because the edges actually in the grand scheme are very sparse it's not an all by all model so this model is extremely more tractable to compute than like an all by all matrix of just estimating the coefficients of how related terms are so it's a very constrained model space but it still describes a lot of system and the second figure adds in this γ sub a which is the precision which is one over Σ the uncertainty so just as we had talked about those being two sides of the coin earlier this is like uh uncertainty slash precision depending on which way you want to think about it on high uncertainty you're really unsure that's low precision and vice versa but it still maps on to the sensory outcome in both these cases after the sensory outcome occurs then there's uh an update the prior becomes the posterior after getting a precision weighted updating and that's just how bayesian updating works so this is a bayesian scheme for updating that initially doesn't include a precision on the a and now it does include a precision on the a any thoughts on two before three yeah i think it's interesting it's good to know that the the expected precision is is tied to the likelihood mapping it doesn't try and tie itself directly to this state it's it's it's in the way that the state is being so that that's that helps to really notice that and make that clear because that gives the logic a little bit more of clarity i think that's often a bit that's not teased out as often as it has been here that's quite useful

awesome cool and what else is interesting so good observation that the uncertainty is connected only to the mapping so it's not uncertainty about the state it's about the mapping in this formulation now we're going to see a similar move and two things happen in figure three to bring that previous model into this uh kind of twice generalized space of generative models of action and perception the first thing is that now there's observations through time versus at a moment with just one observation so subscript one two three through time three time steps then here's that d s a o stack and then b is the transition matrix between states through time so we talked about that on other streams before and then again a similar move just like you pointed out stephen how the the uncertainty or like the intervention in this case the policy π it's actually modulating the transition mappings rather than the states themselves i'm sure that has very interesting computational properties that i don't know the details of but i think that's some components that make it amenable perhaps to message passing or to the variational inference methods but that's definitely a interesting observation that you made it's true it happens again here in a totally different way and then π the policy selection is influenced by preferences which play into the calculation as expected free energy as well as this baseline prior e so we've seen this one before and this is how policies are selected over u for the next time step based upon the u expectations hidden states and observations through time in relationship to priors e how they're u apt to perform before starting this sequence of events and then the way that they calculate their preferences over observation states which also relates to how you said the body centric yeah this this might be a question daniel for them but it says preferences over outcomes and it goes into the expected free energy is that almost like different ways you could weight the way the expected free energy terms presented you know over maybe hamilton's least action versus resolving you know is it would be doing something like that i'm not i'm not sure we can answer that but yeah that makes sense it's a preference for sensory states and then we saw in the model stream one with ryan smith like if you wanted to have a very weak preference in the matrix it would be like 1 versus 1.1 i'm just making those numbers up but the point is like they're relatively close it's like a 60 40 election so that the preference is very weak and then you could also set it up to 100 versus one but then if you think about that like when you're just totally thirsty and you're willing to engage in all kinds of actions that's how you could computationally put in direction as well as magnitude of preferences over outcome some people described you know maybe some individuals have a broad sensory tolerance others have a narrow sensory tolerance or preference so that's so you you've got you so it's almost like you can you can narrow you can either open up or narrow

down the the amounts of options that are going to be available by saying the gap yeah and then in the absence of preference if you don't have a preference for pheromone it's just going to be a diffusion walk so that's how preference plays into the decision making but also other pieces do too okay so then here here's where we introduce this second nested layer with the deep generative model of mental so here the same d s a o is recapitulated from the orange but then the o is split into two nodes one is an estimate of the state but also it's passing or connecting to the precision the uncertainty the features that were being discussed like up until figure three as being just part of how the um inference was performed are now still doing the same exact rule they're just being emitted by the state through the a s sub 1 of 2 like the second level in parentheses through the a matrix at the second level and then that's resulting in these two outputs so this could be like a nested layer of thinking um or a nested layer of concurrent programming or a nested layer of formality dean i'm not i'm going to get the author to make sure that i'm getting this right before something i remember from what kepper talked to with his effective piece and then there's a figure 5 which we'll get to but to me the figures one two and three aspect of this are the unplannable parts of it the figure four that asks the question why am i or yes why am i trying to pay attention to it's kind of the transition space and then the figure five is where we could actually do some planning because we're saying to ourselves now pay attention because we're we're putting idea that second order we're paying ideas to ideas piece so i think up up and up until this point up up until figure four it was unplannable what we're modeling at figure four we start to transition and then in figure five we can actually incorporate a plan if if we so choose if that's part of our policy nice point these are really helpful labels like pie at the first level what am i trying to do so um in the case of the light hitting the receptor and then is it day or night when we introduce policy that would be like should i swivel the cart you know move my sunflower head left or right move my camera left or right under the preference for light now if you didn't have preference it wouldn't matter right if you don't know where you're going it doesn't matter where you go but if you have a preference vector that gets increasingly stringent you're going to pursue those policies that will lead to getting the observations that are in line with your preference now here the second level is so this is again what am i perceiving what am i trying to do that's the motor cart with a visual receptor and then the hidden state again it's just the same exact placement analogously the first layer is what am i paying attention to and the policy is a policy over attentions so that's what makes this model nesting outside of the smaller perceptual level model so blue did you want to say something or no okay then stephen go for it it's treating the precision and it's

somehow merging that with the um to and pushing that through another matrix and probably doing that i know brian smith talks about a cycle of six in terms of when they model they go through six you know it will iterate through six and then the next level will just sample the sixth the sixth the sixth you know the sixth data point so to speak in the in the in the faster data going below um but i'm i'm not sure how how does that work with the two being combined and there's that the two are combined basically to give you the higher o2 is that right i don't know about the six specifically but here you can see like at the first time point of the slower model this is like the hour hand there's like the estimation or inference over states one two and three and everything about them like what's the hidden state and the mapping and the observation which you can then conditionalize the inference on a subset of those variables but that's what the whole model is then it's like you're estimating at time one about time one two and three and then at time two you're also estimating over that trajectory but the slower hand is okay continue so because i think this is where five is where you're getting that d and it's totally the contribution of the paper so they summarize as saying that they make three and this is go the 19 hespetal for a lot of this structure they summarize that they make three novel contributions to parametrically deep active inference models of cognitive control the first contribution is to stipulatively define the tensional and meta-awareness states as the mechanism of opacity control at different hierarchical levels the second is to provide a formal definition of attentional and meta awareness control via an account of higher level policy selection the third is to formally demonstrate the relationship between meta awareness states and the capacity for deliberate attentional control we're going to look at the model in figure 5 and that's still going to be like sort of the abstract skeleton of the model and then there's going to be a simulation results so that was kind of all the background actually to their primary contribution now in figure five we see this explicitly three layer and this is the primary contribution is this model topology and architecture and then the interpretation of the higher order states in terms of the opacity transparency and then eventually a more generalized perspective on mental action bring this all together with a model that does basically the same trick around the second layer as happened around the first layer so these layer one and layer two are the exact same from figure four and now there's this third green layer where the states are asking the unobservable question how aware of my how aware am i of where my attention is and then the policy question is am i trying to maintain awareness of my attentional state or something like that we can get at what exact wording these parameters map onto totally interesting paper and structure of the model in question so any thoughts or what was exciting

to people about figure five stephen and then anyone else yeah it's interesting it's exciting to see it's got some similarities to the valence um the deeply felt effect valence has been a higher level sort of calibration i suppose in a way they might try and find a way to put those two together you know when you say you're i'm aware of where my attention is um well what kind of awareness is that awareness is a it feels good or is it something else so um yeah dean or blue otherwise we can i just want to say a couple quick things so i i the way that they laid the paper out this made a really good case for why they were focusing on the transparency and the opacity opacity piece and then and they don't make any claims around whether or not a person who's looking at a white page with white text written on on it can actually see that white text or whether or not the the size of the font is so large that you're only seeing one pixel in the letter a on that page so you can't even see the a but i think both of those things matter when we're talking about attention as well how can you pay attention to something where you can't see the contrast or how can you pay attention to something where the proportions are so out of whack that they just don't make any sense to you now i know they're not talking about that here but it would seem to me that that would also be a part of where this computational piece would be really really helpful because to a learner sometimes it's easy to see the tunnel within the mountain sometimes it's really really hard to be able to pick something up off the page when it's either way out of proportion or there isn't any contrast so i just wanted to kind of i want to i want to give the authors credit and i think i i hope i think i know where they're going next thanks yeah let's go into a little one cul-de-sac of the formalities that is kind of interesting and then we can also ask the authors to explain a little bit more to us but just to get at how does this s state sub t so through time across time points at the second level so that's going to be what we're focused on is how this is kind of at the intersection of perhaps you could even say the bottom up in the top down signaling or at the very least it's at sort of a um you know mid-level street in the scale of streets of variables from local neighborhoods like down here in the bottom cluster to sort of regional hubs to this broader connection that is happening at an even slower or deeper timescaler way we're kind of looking at this middle piece as a as perhaps an example of again how factors from bottom up and top down interact and then it's kind of like legos maybe it can be generalized from there and you can say okay let's nest this within that or see what happens if you connect over here but they write mathematically the attentional and meta awareness state inference is calculated based upon three sources of evidence red prior beliefs so this is the b and the s at layer two so those are um like what is previously influencing that layer specifically so that's what happened before at that layer direct

perceptual evidence eg the metacognitive observation of one's attentional state blue and then orange ascending evidence based on the precision beliefs of the level below so maybe that's where if you know you just opened your eyes in the morning and it was too bright and you don't have the resolution now you can't make the good mapping between fine-grained details because your resolution from a sensory perspective like the precision is adapted for a different regime of environmental regularities so there's perhaps many ways to model this then they say something that was kind of interesting for the ascending messages from the continuous expected precision to the discrete states we'll use bayesian bayesian model reduction for the derivation see first in 2018 to evaluate the marginal likelihood under the priors associated with each higher level state gives in the case of the attentional evidence for example and then the equation that has these components that were here so it sounds probably extremely technical but it's a pretty interesting topic that we'll look at just in another slide because it is really important to understand how this bottom-up continuous info becomes translated to discretely different states this is just to note that this is like the general structure of the model which you could imagine is also computer variables but then they preset two mental states associated with just specific parameters that they chose especially the simulation is just like a parameter combination it's not all the possible dimensions one can explore so that was just to know so here's oh yeah steven go for it before we go to model i think this is helpful to think about that how that top level has that impact you can imagine you know you wake up sometimes say you're at home and you wake up it's just and slowly you just you see the room you expect to see okay and there's the other times when you wake up and you've gone away and there's and there's that jolt because the prior expectations were still locked into like you're dreaming about waking up at home or you're sort of but you're not home right and there's that so it's almost like you're fulfilling your expectations these these higher level could just start to percolate out as you enter into certain states or into certain rooms um so i think that's quite a nice way of thinking top up top down bottom up and that being about perception as well cool nice point deep priors narrative priors how did i get here with sensory priors um at the moment to moment so here's the paper with fristen parr and zeidman from 2018-19 bayesian model reduction the yellow parts are highlighted but just one green part and then kind of take away here is we consider structure learning and hierarchical or empirical bays that can be regarded as a metaphor for neurobiological processes like abductive reasoning so i know that that's what we're kind of excited about and the paper that we brought up of majeed benny but that's super interesting like just to kind of skip to the last sentence how are they going to be using

model comparison of bayesian models to explore abductive reasoning one takeaway but really would be great to have people who know more about this topic have a little bit more insight would be like we can compare bayesian models that only vary in their priors it's really easy to compare the similar structures of models that just vary in like whether they think it's going to be a gaussian centered at five or centered at seven there's a lot of tools that help you very rapidly compare that but if you have two totally disparate model two programming languages they're predicting different things it's really hard to compare them so the frequentist comparisons can be very intractable as well as sampling and simulation based ones that are just like brute force or they can have stringent requirements like the hierarchical likelihood ratio test is only applicable to certain subsets of frequentist modeling so structure learning is hard what is the structure of causation what's the structure of the data the shape of the data deemed like you brought up so how can we frame this question and challenge about learning about structure of data in terms of model comparison or updating of priors amongst models that have similar structures but just different quantitative variables and so bayesian model reduction is a way that you can do this kind of structured learning and one loose analogy is the lasso regression which tries to restrict to a very small set of few strong factors in a data set that might have like many weakly interacting factors it kind of truncates the distribution just as like give me the most relevant ones and if it's just massively interacting you know so be it and that kind of a regression is gonna have a similar use which is to converge upon a smaller number of things that are really informative while keeping the model also similar which should be reminiscent of like the bayesian information criteria aic as well and then here's in that paper and it relates to what we read earlier how do you go from like quantitative continuous estimates in this case coming from perceptual updating of like that retina sensor how do we how do we plug in from the continuous and map to categorical discrete selection of models so mapping from continuous to continuous you can just do like a dilation or some other function mapping but it gets a little bit different when you want to map between these worlds of continuous and discrete and it turns out that the way it works is that inferences performed within and among a family of models bayesian models that are using parameters like quantitative difference in parameters to describe structural connections between variables so here's like let's just say that black is zero and y is one then here's like all the combinations of linear models that were used to generate the different models so this is the exhaustive frequentist approach here what's happening is the bayesian model is estimating the relevance of each of up to 20 parameters estimating like a credible interval which is 90 percent in this

case 90 of the time it's within this interval we think or that's where it's likely to be under some value of stringency that's the red blur and then you prune the ones that are like not significantly different from zero so it doesn't mean that they're not zero or they're not playing a role or the proteins don't touch but it means that you can actually structurally simplify the relationship between variables so that's moving from continuous evidence about models of a discrete family to a reduced model that could potentially be extremely parsimonious so that's just the bayesian model reduction but that's like i guess a sub component of this paper so wanting to know more about how it comes into play or if there's an analogous process on the generative way down okay any thoughts on that or just that was one little section well i was just thinking i mean i'm not sure if i've got this potentially right but i can see this brings in the idea of how action can be used to think like in creative abductive work you you know you can start to actually create entropy or create confusion to create some possible outcomes and that the art of the creative process isn't necessarily the creation of random novel ideas it's been able to prune back i think it comes a bit back to the idea of that task related off-focus mind wandering you know the ability to then go backwards and forwards and that might tie in with this ability here to do this i'm i'm not sure but it sounds like this would tie into that somehow cool we'll ask the authors okay nice onto the simulations also people can just leave when they need to this section is going to provide the simulation results that show how the model described in the previous section engenders opacity transparency phenomenology so outputs of the model that recapitulate some of the dynamics that have been described in other experiments and then they're gonna use that agent with a three layer generative model okay so first was just um uh well any okay any thoughts on like what the simulation is or just to contextualize it before we talk more about it the code is available it's a collab book it's pretty cool it it uh makes a lot of graphs so nice work for the authors maybe we can walk through it or ask about certain parts but cool that it's a interactive notebook anyone want to describe seven sure i will so here it says um just the yellow part says in the first half of the trial the agent is in the focus state which confers a higher precision i.e the agent is generally more confident about how their observations map onto states in the second half of the trial the precision drops as the agent moves into the distracted state as a result their beliefs about latent states are updated more slowly so this is kind of like i wasn't really paying attention did i just see a car flash in front of me i don't really know because i wasn't really paying attention um whereas if you're focused on the screen and watching the car you see every flash and you know exactly when it flashed way to put it blue so the blue is what um happened it's like a

tone like a lower piece even like a lower tone versus a higher tone that happens some frequency of the time and then um it is just this is a agent that's artificially held with a total focus state for the lighter half and then it just switches instantly and we're exploring how this s at the first level so to go back to figure five that's the perception is s at the first level and then that is an estimate that's changing through time and so um when it's focused it's like higher highs and sharper and lower lows and then distracted states just more like wait what stuff's more just spread out and it maybe has a different recovery or similar speed but lower to rise so that's how they say that that's how they make their claims about the way that the attentional state influences a1 by making the likelihood mapping more or less informative because if it's uninformative the likelihood mapping why does the sensory input matter go with your gut but if it's very precise you want to go with what your lying guys are telling you and that is reflecting in figure eight this um phenomenological cycle so these can be discrete just via that mechanism of like the continuous to discrete estimation and they write that this is the diagram of the phenomenological cycle that occurs during sustained attention tasks the process cycles through being focused becoming distracted becoming aware of this distraction and then refocusing so it's cool because it's like it looks like internal external states in a way and action and sense states like it's it's a four it's an ooda loop or there's so many other four-fold models but what are we actually looking at here we're looking at the dynamics of that attractor state as it wanders i feel like since both of you thought about attention or awareness maybe what did this like look like or what does this mean to you do you think that we we attend through steps all the time i know that for the simulation they want to be able to see that in order to be able to do the calculations but do you think people do so i think that the shift from focus to distracted is maybe not discrete like that might be like us like a more of a continuous um transition but definitely like as soon as i'm aware that i'm distracted like the attention gets redirected like that is a the awareness of distraction is a discrete like bam oh um and then the redirection of attention is is also like i can replace it and then i'm focused again but then the transition so it's just this top left corner where it's a little fuzzy for me i i really agree with that blue like awareness of distraction to redirection of attention that's like you know the eye of mordor it just goes somewhere else okay but then the distracted to the awareness of distraction is actually that same but it's the eye the cognitive metacognitive eye of mordor turning its eye on a lower level estimate so i agree that actually this from from from noon to six it looks it's pretty interesting how it maps but do we think about focus and distraction as part of a continuum of a single variable or is this like a yeah is there can you wobble or is there like

more dimensions to just being super strongly obsessively doing something how far do you pull back what if you have to you know put down the first pen to pick up a different pen you're pulling back from the goal but you're moving towards it so that's the kind of deep policy questions that are so cool that we're raising blue well so it's cool because um here it's this gamma function right like so the gamma is the precision and so just by changing the gamma and like you can change the gamma incrementally however you want to change the gamma like 0.01 0.1 0.001 and then by changing that gamma you can really get that shift like computationally from focus to distracted by by just by changing that precision metric yep and i mean that's kind of imagine the little headband when people are paying attention and what do you do so here i copied the left side of i think five and then just laid it next to here so these are the questions that again each layer is asking the blue layer is like the perception layer and related to the policy that's going to get those perceptions in line with preferences so moving the cart back and forth to be in the right amount of light then hidden mental states are about paying attention what influences the precision and emits a precision variable and an observable at a higher level just like a sub one at the second level here's like a_3 and here's now the slow one but it's happening actually at every time point so again you could probably elaborate this in various ways and ask like how many time steps of action are calculated per how many attentional states are calculated per metacognitive state are they all in the flow it's so so here then they also put down the specific matrices i assume the ones that we'll see in this colab notebook these are just like the parameters that are going to illustrate some of the dynamics that are going to be interesting for the fields but again these are parameters that could be learned or updated or could reflect not just a two by two option all right so maybe even copying this guy out again here figure yes okay so here are the three layers of the model and so the bottom layer is playing the oddball paradigm so it's like listening to like regular tones and then once in every x percent there's a differential tone but also that could be like which one of these security events is anomalous or which one of these trades would be effective amidst the prior that 90 are probably not it starts off focused and the sense all systems are normal on the sensory side and it gets distracted it returns to being focused this is the true state at each level is the dot so the bottom layer is like the percept of the sound this is the attention this is the meta awareness and then this you see at the change point from being distracted to focus there's a switch point in the third level so then everything's normal the sensory level and then it's distracted and then the the oddball happens at 21 or something and then that like flips it it says flips it back into attention and then that continues and it stays in

that state until an event happens and so this is just simulating out the dynamics but the idea with this kind of a graph is that we can scan along and look at how changes in one level either are influencing each other statistically causally within our model not necessarily about the real world but just literally causally because we specified it in our model and then do parameter fitting and be like whoa we can explain 95 percent of the variance in this behavioral data set even though we didn't observe the red or the orange we observed this or their task variability but then we have enough data points from enough individuals and from time series of this individual and then start to add in some other measurements and there's a lot of data to do inference on these higher order states about any comments on 10 i'm just thinking so sorry i'm just thinking about um how this is going to be used like by companies to keep you working like i just can't help but like think about about thinking about that like are you paying attention are you not paying attention can we give you an oddball to make you pay attention like you're sitting at work and all of a sudden like oh i wasn't even paying attention so um yeah like that i think about that and it's kind of scary and it's kind of um i don't know it's real i think blue if people actually tried to do that i think they would be more misinformed than they would be informed as i said for half a dozen years i worked with people who whose goal was to pull apart and sort of get a better understanding of this time on task piece and they're no closer today to understanding it they were a decade ago and they spent a lot of time on it because i don't know that this literally maps into an appreciation or understanding of what on task versus off task is there's tangents to tasking and i don't think that this really gives us a sense of what that is i think the blue layer does speak to the unplannable i think that the middle layer does speak to transition and i think the top green red does speak to planable but if people if somebody in an organization thinks that they're going to have a better sense of how they're going to determine whether a person was on task or not i say good luck with that so i actually have a colleague that's developing a way to to determine if a person's in the flow state or not which is a very different very interesting like with physiological so attentional is different like are you paying attention versus are you are you in flow and it's it's really interesting and they've got some some decent results so far yeah well they can they can count the numbers saccades or saccades and i've looked at the whole visual memory thing and all of that literature and it's still not it's still they're still not getting any closer to understanding when a person is on or off task because they want to break it down to that kind of a binary i say let them continue to let them to continue to do that because i i i'm not sure if it's nefarious but i think it has the potential to become that so i'd like them to go and spend all their time on it because they're not

going to figure it out keep them distracted oh dean laying down breadcrumbs i love it carry on figure nine is gonna be that same cycle of four that we were talking about on eight and now in addition to just the sort of qualitative label we're going to map that onto the uh structure of the model that we were just discussing so here like the redirection of is being framed as a policy to π at the second level where's the action π at the second level is this orange one here on a figure five so that is influencing what one is paying attention to through time and then that is being mapped to here and similarly you can read through and just work through the combinatorics here of what these map to from a dynamical systems perspective like while it's in a parameter neighborhood or regime or attractor or manifold that's in one area in a distracted state described by a wide range of continuous parameters that map onto the categorical distinction of the true state being distracted but the perception being that one is focused so just work through what these distinctions are is pretty interesting i yeah anything to add on that it's kind of cool um 11 is gonna be another kind of just like in seven they had the attention at the second layer at one parameter level and then they just like turn down the dimmer in figure they're gonna have a high meta awareness and then turn down the dimmer to the low meta awareness at the s3 so recapitulating the way that they added this third layer to hespedale 2021 they're recapitulating figure 7 in how they vary the parameters changing but now you can see the dynamics are totally different like the spiking between focus and distraction is actually happening faster during the period of fixed high meta awareness whereas during the period of low meta awareness actually it's like there's longer intervals it's like well what just happened i was having low meta awareness the time just flew by or maybe there's another interpretation but it's just um a nice plot that gets at how just a parameter combination can give us insight into the structure of the model so we kind of have the graphical doorway the programming doorway and then this is like the outcome oriented data science way because this is just input in a model it's just in a machine learning notebook it doesn't really even explicitly invoke the graphical structure that we just have been talking about the whole time but it is that embodied in a simulation all right dean nah i gotta talk to the author tomorrow because so it brings up the thing about is it tomorrow yeah i'm going to bring up what i what i used to talk about as the crosswalk analogy i'm just going to leave it at i can see you being a great crosswalk guard you wait figure 12 this is kind of the last figure i thought this was an awesome figure and one i i know that we'll return to many times a base graph of a deep generative model of mental action generalized over all precision parameters so now that attentional state the likelihood mapping is going from potentially binary state like we

saw here focused or not i mean uh yeah um the likelihood mapping kind of like the dark side of the moon album cover can emit like five simultaneous outputs it can output a tuple or a vector from a computational perspective perspective and it can output what we talked about but it can also emit an uncertainty and potentially even other things connecting to multiple of these lower level parameters so now there can be an uncertainty over the priors whereas in the previous simulations d just stood alone and there wasn't uncertainty around it so it's just sort of taking the next step and just being like okay if we um went from estimating the valence with the actual like plus or minus are things going better or worse than expected and we just said okay what's the sister of valence reward the sister is the uncertainty and the precision and their relationship so now this paper that's why it builds on hesp 2021 because it fills in the other complement there which is the precision modulation by higher level states rather than the valence modulation by higher level states how well are things going and then that insight to generalize into the variation from a top-down level has strong precedent in bayesian modeling where it's related to like the parametric empirical base and hierarchical base big a lot of philosophical questions about what a parameter means challenges in implementing it but this is a relatively generalizing step in the publications um oh yeah exactly that it's the same higher level affect of states applied to likelihood precision rather than model precision note that this evidence is only available when higher level likelihood precisions is non-zero i.e when the state is opaque to some degree as a result the transparency opacity distinction can be directly related to the evidence calculation so what does that mean or just interesting like we don't notice something and then we do so it's kind of like it's transparent and then it becomes opaque and then we observe it in this model it puts us over threshold of observation that takes us off task as defined by okay on to the implications and bigger questions so blue go for it well so just to your last point like i wonder if um you know have you ever felt so focused you're distracted like like that's that's a common thing for me um and maybe it's focused on the wrong task but but it's also like you're focused to the exclusion of of everything else so i just wonder how that goes into this transparency opacity gradient maybe so so anyway and then um yeah on to uh implications and bigger questions so i i think about oh sorry go ahead did you have something some response dean oh oh go for blue uh so yeah so uh implications and bigger questions so here i like when so i obviously got one during this discussion about being so focused that you're that you're not paying attention but also uh if we're we're talking about adding hierarchical levels so i wonder about here beyond meta awareness um i wonder about collective consciousness like so is there you know when you realize that you're part of a

collective and then when the collective develops some kind of meta consciousness like so so it makes me think about about that and about maybe how to model something like that maybe an ants for example because there's there's the individual level and then there's definitely like the society structural level the consciousness of the colony maybe i mean also uh you know simulation theory like so so at what level like becoming aware that we're in a simulation like like where does that fit into the hierarchical nesting of this deep um you know meta funny stuff wouldn't have thought that that's where we would get you know but that's kind of what it has come to any thoughts on that team before we go to the next slide no i think it ties in nicely with the what are we metacognet metacognitizeober yep and i'll i'm going to bring up the crossblock tomorrow yep so this future of phenomenology phenomenology just whatever we want to call it um here's just a few angles on what is slash has and will happened there's multimodal brain measurements ospm where art they sting but just it lays out in 2007 and earlier but in that textbook and in the online documentation kind of generalized frameworks for harmonizing across different modalities of neuroimaging behavioral imaging hidden state modeling so a lot of the seeds are in spm what we can talk about integrating now other than in a laboratory setting like eeg and fmri at the same time together but increasingly you know even at home physiological sensors wearables cameras and environmental sensors the capacity for models that involve multiple humans and finding the synchronies and attunements amongst them under environmental changes and this is like the work of hyper scanning and guillaume de ma and other a lot of other people in that field thinking about cyber physical systems like what if the state is inside the head purportedly of a human is that different than if this state is purportedly inside of a software which ones get to be affect which ones get to be what standing and then also it helps us and always is important maybe you'll go here with uh who and why and how dean but really just all these axes that phenomenology has been addressed from developmental multicultural neurodiversity a lot of historical perspectives that also is continuing to weave together and not being said to be obsolete or simplified by a computational model yeah if you want to know more phenomenology go and reach sean gallagher nice don't know about that but it sounds cool then yeah anything there or lost okay then here was just one paper that kind of came out just on a closing note like how it connects out maybe to other areas this was looking at behaviors across a bunch of different fruit fly species drosophila here's the big figure or one of them kind of mapping just like you might map genes being correlated or might map neural activations to be correlated across different regions of the brain those same matrix methods tensor methods fundamentally covariance

estimation can be used to study different kinds of relationships so earlier we saw like the neural activity in the behavior here the heat map is like behavior to behavior they're looking for pure behavioral covariation but then what's really cool about this paper is that they use several techniques from information theory which we see all the time in fep information thermo cool going that way but then also puts it explicitly in a comparative evolutionary framework so this is really moving towards models that can just very seamlessly integrate different kinds of data as easy as integrating the same kind of data just the matrix is type a of data by type b of data rather than type a by type a so these kinds of models that cross data input types need to be connected to broader traditions like thinking about development ecology evolution it's just also interesting how these fields which are increasingly data rich are going to be using different cutting edge theoretical frameworks so that just one closing thought about where is evolution in this where's developments where's what are we doing all these components discussion elicits any last thoughts otherwise thanks blue and steven and dean because this was fun to work on and i think we all got to learn a lot more tomorrow yep i know i know it's only a few we've just been so not on top okay all right thank you again thanks everyone for watching peace out bye